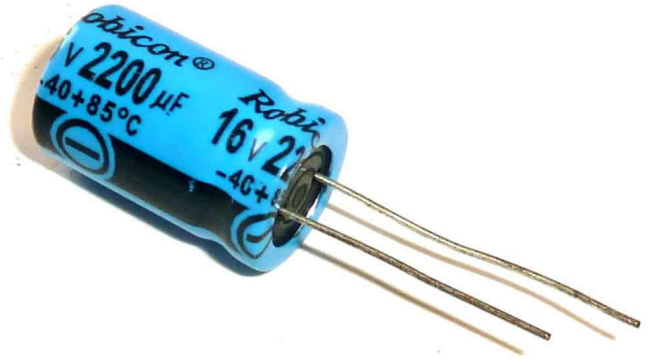
Capacitors #1



Capacitors

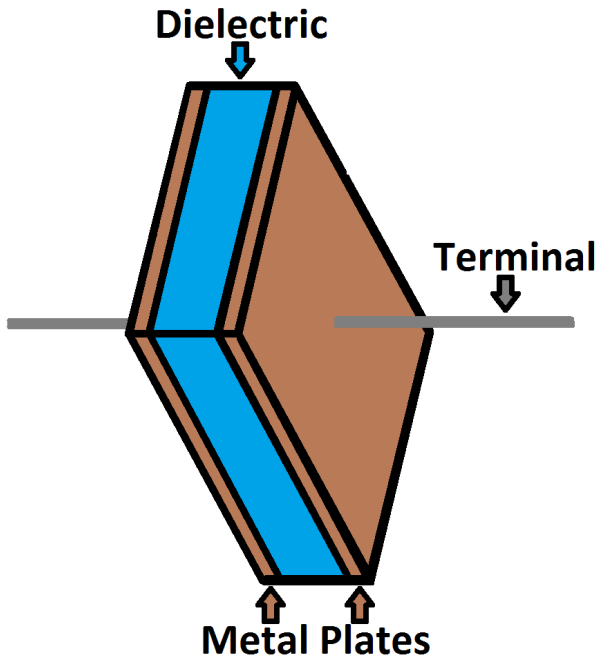
- A **capacitor** is an electronic component that stores and releases electrical energy.

- Capacitors are commonly used in electronic circuits for functions such as:
  - **Storing energy** (like a small battery).
  - **Smoothing voltage fluctuations** in power supplies.
  - **Filtering signals** in audio and radio systems.
  - **Blocking direct current (DC)** while allowing alternating current (AC) to pass.



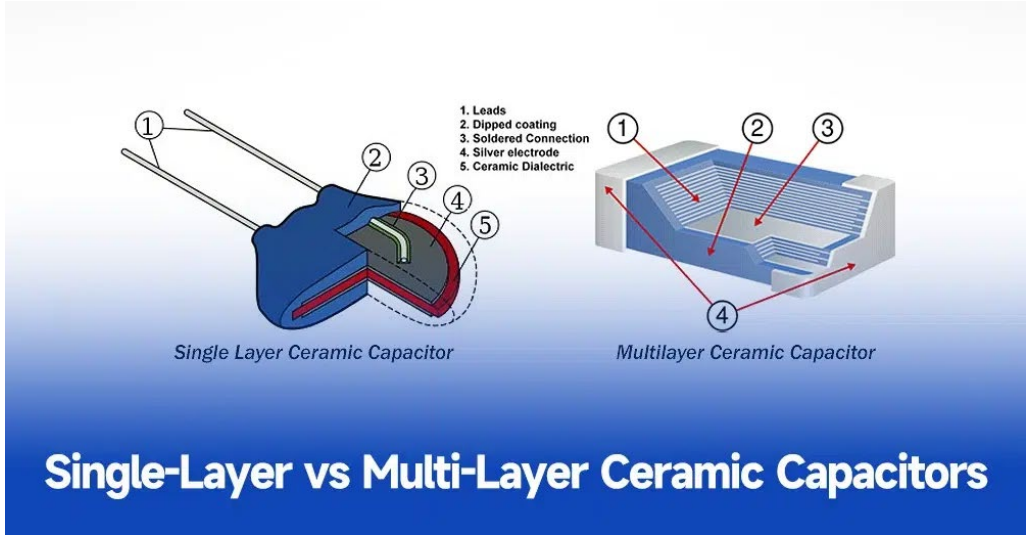
What is a capacitor made from?

- A capacitor is built from 2 plates made from conductive material
- These two plates store and create an electric field
- In between the conductive plates there is dielectric material, this material acts as an insulator between the plates
- This insulator gives a higher capacitance value



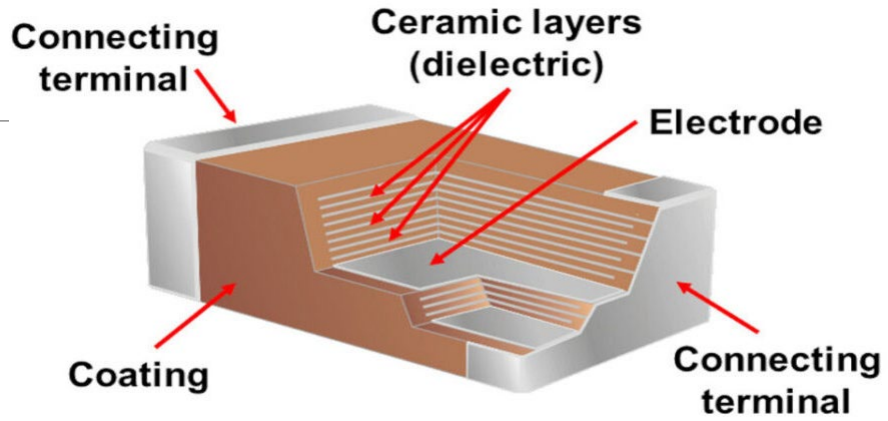
Types of capacitors

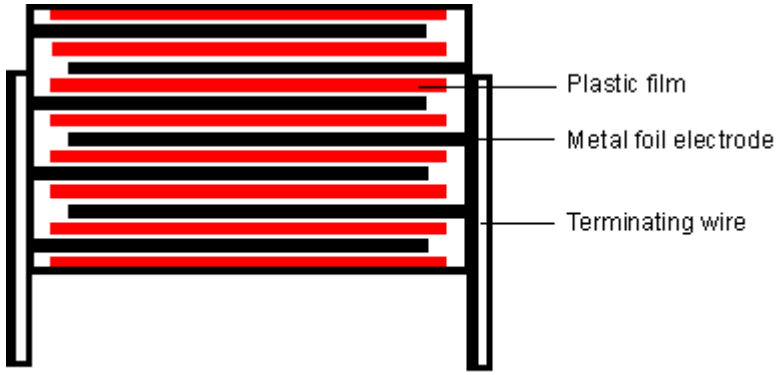
- There are two ways of constructing a capacitor:
- Multilayer Capacitors (MLCs) feature multiple layers of dielectric and metal
- Single-layer Capacitors feature a single layer of dielectric between two metal sheets



Dielectric Materials

- The dielectric dictates the capacitor's values as well as how it functions
- A dielectric must have insulating properties; however, it can be many different materials
- These include:
  - Mica
  - Plastic
  - Paper
  - Ceramic

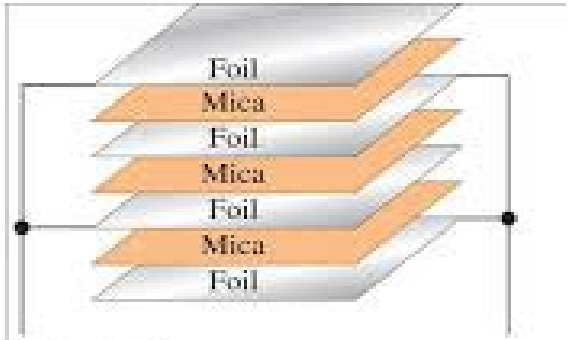




Dielectric Materials - Mica



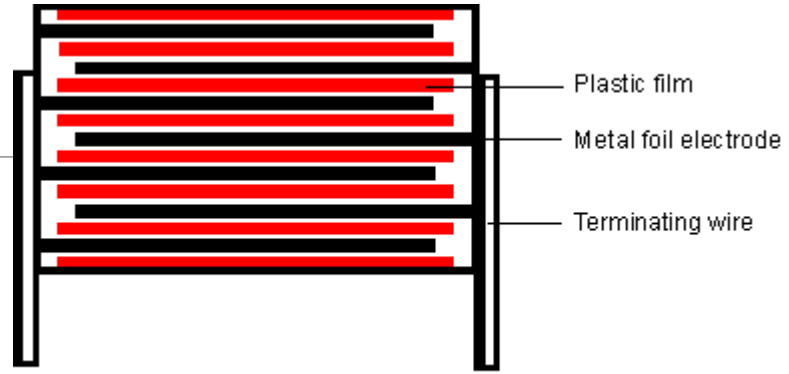
- Made from thin sheets of mica – a naturally occurring mineral
- These sheets are stacked and layered.
- Mica's stability ensures a consistent and durable dielectric.
- High stability, low loss and high precision are all characteristics of Mica



(a) Stacked layer arrangement

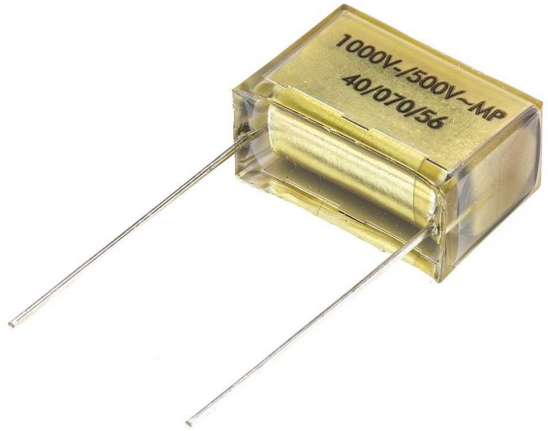

(b) Layers are pressed together and encapsulated.

Dielectric Materials - Plastic



- Made from thin sheets of plastic film, including:
  - Polypropylene (PP)
  - Polyester (PET)
  - Polytetrafluoroethylene (PTFE)
  - Polycarbonate (PC)
- These sheets have the metal deposited on them forming the capacitor
- Plastic can be easily formed and rolled/stacked making production easier
- High stability, low loss and a wide range of capacitance are all characteristics of Plastics

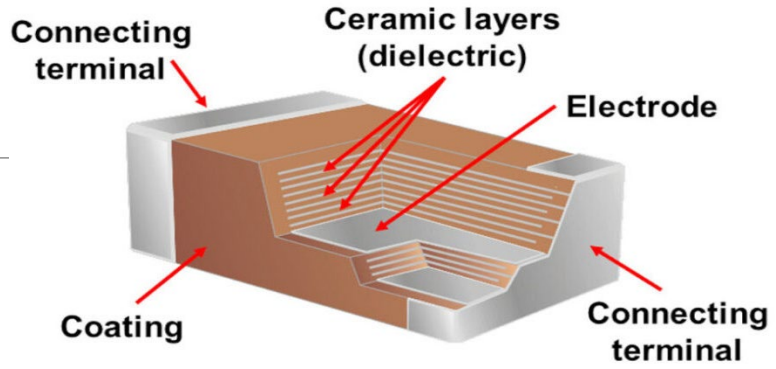
Dielectric Materials - Paper



- Made from thin sheets of paper in between metal foil
- These sheets are often impregnated with oil, wax, or resin to enhance its insulating properties and protect it
- Paper is relatively cheap and easy to create compared to other dielectrics making it one of the first dielectrics used
- A low capacitance/voltage rating, moisture sensitivity and dielectric loss are characteristics of paper

Dielectric Materials - Ceramic

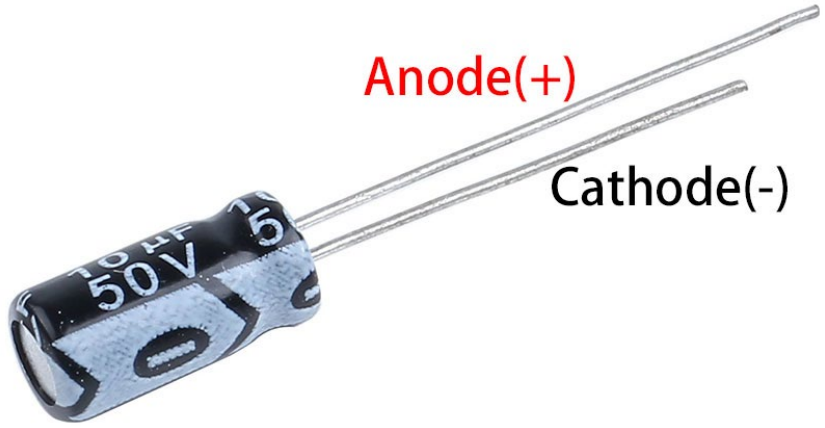
- Made from thin layers of ceramic including:
  - Barium Titanate and Titanium Dioxide
- These sheets are layered between metal sheets
- A high range of capacitance, good temperature regulation and the ability to handle high frequency are characteristics of ceramic
- Ceramic capacitors frequently are used as single layer capacitors





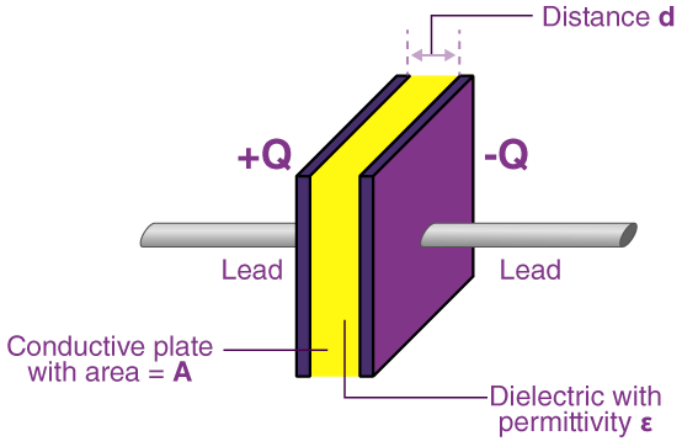
Anode and Cathode

- There are two main terminals on a capacitor, the anode and the cathode
- The anode stores positive charge
- The cathode completes the circuit and stores negative charge
- In most capacitors the anode and cathode are determined by how its plugged in
- However electrolytic capacitors have a set anode and cathode



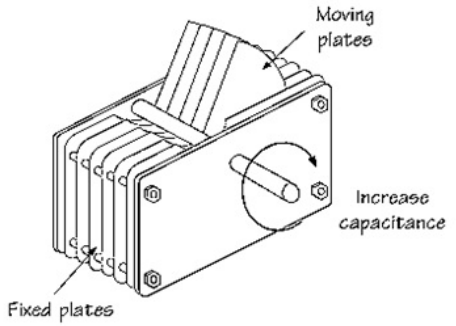
Equations for capacitance of single capacitor

- **Capacitance** of a parallel plate capacitor:
  - $C = \frac{\varepsilon_0 * A}{d}$
- **Capacitance** of a parallel plate capacitor with dielectric:
  - $C = k\frac{\varepsilon_0 * A}{d}$
- ε_0 = Permittivity of free space = 8.85×10^{-12}
- k = Dielectric Constant



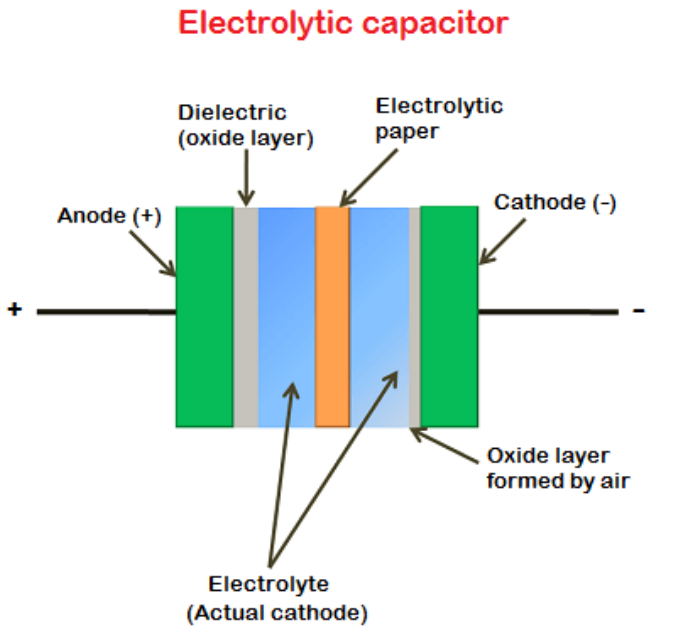
Other types of capacitors - Variable

- A variable capacitor can change its capacitance using mechanical movement
- It typically has air as a dielectric as it easily allows movement
- It's built out of two parts, the rotor and the stator. The stator plates remain stationary and the rotor parts turn.
- These rotors and stationary plates play the role of the electrode plates



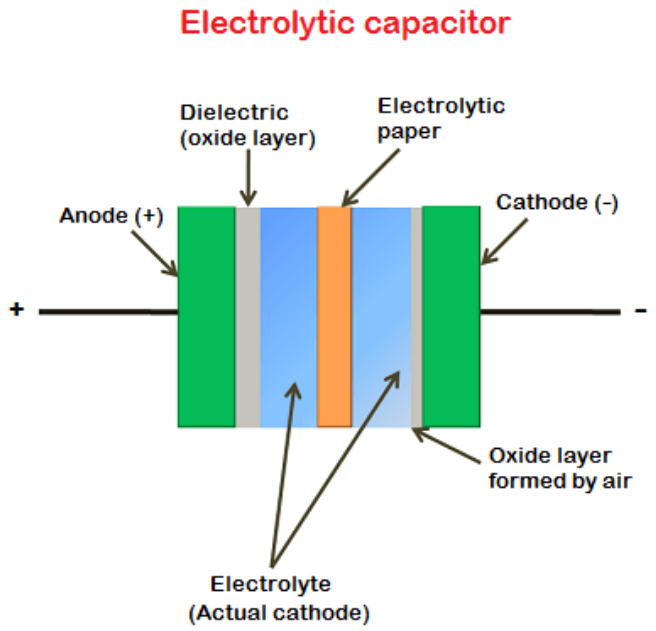
Other types of capacitors - Electrolytic

- An electrolytic capacitor works like most other capacitors with a metal anode and cathode and a dielectric
- However, it also has electrolytic solution which is referred to as the “actual cathode”. The electrolytic solution acts as the cathode, it just can’t easily transfer out the current
- The metal cathode just collects the electric current from the electrolyte to complete the circuit
- The dielectric is a thin oxide layer on the metal formed by air oxidising the material



Other types of capacitors - Electrolytic

- Electrolytic capacitors have etched anodes and cathodes; this increases the surface area which the electrolyte can contact
- This higher surface area makes a more efficient capacitor as the electrolyte can get into every nook and cranny
- They can only go in one direction as the oxide layer relies on the chemistry of the capacitor, if the electricity goes the opposite way it destroys the layer and shorts the capacitor



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Capacitance

- **Capacitance** is the ability of a capacitor to store electrical charge.
- It is measured in **farads** (F), typically expressed in smaller units, such as microfarads (μF) or millifarads (mF).

$$C = \frac{Q}{V}$$

- C: Capacitance, measured in **farads (F)**.
- Q: Charge in coulombs (C).
- V: Voltage in volts (V).

$$1\ \mathrm{F} = \frac{1\ \mathrm{C}}{1\ \mathrm{V}}$$